

Multivariable Calculus (MT1008)

BSAI, BSDS, BSSE (All Sections)

Course Instructors

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Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 50

Total Questions: 4

Date: February 25, 2026

Roll No

Section

Student Signature

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Attempt all the questions

[CLO #1: Apply multivariable calculus concepts to solve standard mathematical problems involving partial derivatives, multiple integrals, and basic vector fields.]

Question 1

[8 Marks]

- (a) In spherical coordinates, the coordinate surface $\phi = c$ changes its geometry as ϕ varies from 0 to π . Describe for what value(s) of the c the surface represents the positive z -axis and a downward-opening cone. (2)
- (b) An AI system uses a similarity score S to compare two data points in a 2D feature space. The score is defined as:

$$S(x, y) = \frac{1}{\sqrt{16 - x^2 - y^2}}$$

- (i) Find and sketch the domain of the similarity function $S(x, y)$. (2)
- (ii) Is the domain D an open or closed region? Is the domain D a bounded or unbounded region? Justify your answer. (2)
- (iii) Describe the interior and boundary of the domain. (1)
- (iv) For a fixed similarity score $S = c$, determine the equation of the level curves. What geometric shapes are these level curves? (1)

Solution

- (a) In spherical coordinates, ϕ is the angle measured from the positive z -axis.

- When $\phi = 0$, all points lie along the positive z -axis. Thus, the surface represents the **positive z -axis** when

$$c = 0.$$

- When $\phi = c$, where $0 < c < \pi$, the surface represents a cone with vertex at the origin.

In particular, for

$$0 < c < \frac{\pi}{2},$$

the cone opens upward, and for

$$\frac{\pi}{2} < c < \pi,$$

the cone opens downward.

Thus, the surface represents a **downward-opening cone** when

$$\frac{\pi}{2} < c < \pi.$$

(b) (i) The similarity function is

$$S(x, y) = \frac{1}{\sqrt{16 - x^2 - y^2}}.$$

For $S(x, y)$ to be defined, the expression under the square root must be positive:

$$16 - x^2 - y^2 > 0.$$

Rewriting,

$$x^2 + y^2 < 16.$$

Hence, the domain is the interior of the circle of radius 4 centered at the origin.

Sketch: A circle of radius 4 centered at $(0, 0)$, with the interior shaded (boundary not included).

(ii) The domain $D = \{(x, y) : x^2 + y^2 < 16\}$.

Since the inequality is strict ($<$), the boundary $x^2 + y^2 = 16$ is not included. Therefore, D is an **open region**.

Also, since $x^2 + y^2 < 16$ restricts all points to lie within a circle of finite radius 4, the region is **bounded**.

(iii) The **interior** of D is all points satisfying

$$x^2 + y^2 < 16.$$

The **boundary** of D is the circle

$$x^2 + y^2 = 16.$$

(iv) For a fixed similarity score $S = c$,

$$\frac{1}{\sqrt{16 - x^2 - y^2}} = c.$$

Solving:

$$\sqrt{16 - x^2 - y^2} = \frac{1}{c},$$

$$16 - x^2 - y^2 = \frac{1}{c^2},$$

$$x^2 + y^2 = 16 - \frac{1}{c^2}.$$

Thus, the level curves are circles centered at the origin with radius

$$r = \sqrt{16 - \frac{1}{c^2}}.$$

Hence, the level curves are **circles** (provided $16 - \frac{1}{c^2} > 0$).

[CLO #1: Apply multivariable calculus concepts to solve standard mathematical problems involving partial derivatives, multiple integrals, and basic vector fields.]

Question 2

[14 Marks]

An autonomous AI-powered satellite node in a distributed space communication network is located at $P(3, 1, 5)$. It transmits an encrypted data stream toward a ground-based AI relay server located at $Q(5, 2, 3)$. During transmission, the signal may interact with a high-altitude atmospheric monitoring layer used for anomaly detection. This intelligent monitoring layer is modeled as a plane in $\mathbb{R}^3$ passing through $(1, 2, 4)$ with normal vector $\vec{n} = \langle 2, -1, 2 \rangle$. In this AI-driven spatial simulation:

- Points represent network nodes in $\mathbb{R}^3$.
 - The line segment PQ represents the transmission path of the data beam.
 - The plane represents an AI-based environmental detection boundary.
- (a) Write the equation of the straight-line transmission path of PQ . (2)
- (b) Find the angle between the beam and each coordinate axis using direction cosines. (3)
- (c) Write the equation of the detection boundary (plane). (2)
- (d) Calculate the ℓ_2 -norm of the normal vector and the ℓ_∞ -norm of the beam direction vector. (2)
- (e) Calculate the shortest distance from P to the detection boundary. (2)
- (f) A rogue data packet is detected at $S(0, 4, 3)$. Calculate the distance from S to the transmission path. (2)
- (g) Determine whether the data beam will penetrate (intersect) the atmospheric layer or travel parallel to it. Justify using vector reasoning. (1)

Solution

- (a) The transmission path is the line passing through $P(3, 1, 5)$ and $Q(5, 2, 3)$.

Direction vector:

$$\vec{PQ} = Q - P = (5 - 3, 2 - 1, 3 - 5) = (2, 1, -2).$$

Hence, the vector equation of the line is:

$$\vec{r}(t) = (3, 1, 5) + t(2, 1, -2).$$

Parametric form:

$$x = 3 + 2t, \quad y = 1 + t, \quad z = 5 - 2t.$$

(b) The direction vector is $\vec{d} = (2, 1, -2)$.

Magnitude:

$$|\vec{d}| = \sqrt{2^2 + 1^2 + (-2)^2} = \sqrt{4 + 1 + 4} = 3.$$

Direction cosines:

$$\cos \alpha = \frac{2}{3}, \quad \cos \beta = \frac{1}{3}, \quad \cos \gamma = \frac{-2}{3}.$$

Thus, the angles with the coordinate axes are:

$$\alpha = \cos^{-1}\left(\frac{2}{3}\right), \quad \beta = \cos^{-1}\left(\frac{1}{3}\right), \quad \gamma = \cos^{-1}\left(\frac{-2}{3}\right).$$

(c) The plane passes through $(1, 2, 4)$ with normal vector $\vec{n} = (2, -1, 2)$.

Using pointnormal form:

$$2(x - 1) - 1(y - 2) + 2(z - 4) = 0.$$

Expanding:

$$2x - 2 - y + 2 + 2z - 8 = 0,$$

$$2x - y + 2z - 8 = 0.$$

(d) ℓ_2 -norm of the normal vector:

$$\|\vec{n}\|_2 = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{4 + 1 + 4} = 3.$$

ℓ_∞ -norm of beam direction vector $(2, 1, -2)$:

$$\|\vec{d}\|_\infty = \max\{|2|, |1|, |-2|\} = 2.$$

(e) Distance from $P(3, 1, 5)$ to the plane:

$$\begin{aligned} D &= \frac{|2(3) - 1(1) + 2(5) - 8|}{\sqrt{2^2 + (-1)^2 + 2^2}} \\ &= \frac{|6 - 1 + 10 - 8|}{3} = \frac{7}{3}. \end{aligned}$$

(f) Distance from $S(0, 4, 3)$ to the line:

$$\vec{PS} = S - P = (-3, 3, -2).$$

$$\vec{d} = (2, 1, -2).$$

Cross product:

$$\vec{PS} \times \vec{d} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -3 & 3 & -2 \\ 2 & 1 & -2 \end{vmatrix} = (-4, -10, -9).$$

Magnitude:

$$|\vec{PS} \times \vec{d}| = \sqrt{(-4)^2 + (-10)^2 + (-9)^2} = \sqrt{16 + 100 + 81} = \sqrt{197}.$$

Distance:

$$D = \frac{|\vec{PS} \times \vec{d}|}{|\vec{d}|} = \frac{\sqrt{197}}{3}.$$

(g) To determine if the beam intersects the plane:

Compute $\vec{d} \cdot \vec{n}$:

$$(2, 1, -2) \cdot (2, -1, 2) = 4 - 1 - 4 = -1.$$

Since $\vec{d} \cdot \vec{n} \neq 0$, the line is not parallel to the plane.

Therefore, the beam **intersects (penetrates)** the atmospheric layer.

[CLO #1: Apply multivariable calculus concepts to solve standard mathematical problems involving partial derivatives, multiple integrals, and basic vector fields.]

Question 3

[18 Marks]

(a) In an image processing algorithm, the brightness of a grayscale image is modeled by

$$f(x, y) = 10 - x^2 + y^2 + 3 \cos(x),$$

where x and y are pixel coordinates. An edge or gradient is detected by analyzing how this intensity changes at a specific pixel $P(0, 2)$.

- (i) Calculate the exact brightness intensity at pixel P i.e., $f(0, 2)$. (1)
- (ii) Illustrate the local linearization by finding the equation of the tangent plane at the point P . (3)
- (iii) If a scanning window moves across the pixel in the direction $\vec{v} = \langle 3, 4 \rangle$, calculate the rate at which brightness changes. (2)
- (iv) Determine the direction of maximum increase and compute the maximum rate of change at P . (2)

(b) Consider the function

$$f(x, y) = \begin{cases} \frac{xy^2}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

- (i) Using the limit definition of partial derivatives, calculate $\frac{\partial f}{\partial x}(0, 0)$. (3)
 - (ii) Use polar coordinates to evaluate $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$. (3)
 - (iii) Comment about continuity of the function $f(x, y)$ at origin. (2)
- (c) Mention the conditions on the first order partial derivatives that ensures the existence of the partial derivatives at a point (a, b) , and geometrically, what does it mean for a function of two variables to actually be differentiable at a point? (2)

Solution

(a) Given

$$f(x, y) = 10 - x^2 + y^2 + 3 \cos x, \quad P(0, 2).$$

(i)

$$f(0, 2) = 10 - 0^2 + 2^2 + 3 \cos(0) = 10 + 4 + 3(1) = 17.$$

(ii) First compute partial derivatives:

$$f_x = -2x - 3 \sin x, \quad f_y = 2y.$$

At $(0, 2)$:

$$f_x(0, 2) = 0, \quad f_y(0, 2) = 4.$$

Tangent plane formula:

$$z - f(0, 2) = f_x(0, 2)(x - 0) + f_y(0, 2)(y - 2).$$

$$z - 17 = 0(x) + 4(y - 2),$$

$$z = 17 + 4(y - 2) = 9 + 4y.$$

(iii) Direction vector $\vec{v} = \langle 3, 4 \rangle$.

Unit vector:

$$\vec{u} = \frac{1}{5}(3, 4).$$

Gradient:

$$\nabla f(0, 2) = (0, 4).$$

Directional derivative:

$$D_{\vec{u}}f = \nabla f \cdot \vec{u} = (0, 4) \cdot \left(\frac{3}{5}, \frac{4}{5} \right) = \frac{16}{5}.$$

(iv) Maximum increase occurs in direction of gradient:

$$\text{Direction} = \nabla f(0, 2) = (0, 4).$$

Unit direction:

$$(0, 1).$$

Maximum rate:

$$|\nabla f(0, 2)| = 4.$$

(b) Consider

$$f(x, y) = \begin{cases} \frac{xy^2}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (0, 0). \end{cases}$$

(i) Using definition:

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h}.$$

Since $f(h, 0) = 0$,

$$= \lim_{h \rightarrow 0} \frac{0}{h} = 0.$$

(ii) Use polar coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta.$$

$$f(x, y) = \frac{(r \cos \theta)(r^2 \sin^2 \theta)}{r^2} = r \cos \theta \sin^2 \theta.$$

As $r \rightarrow 0$,

$$f(x, y) \rightarrow 0.$$

Thus,

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0.$$

(iii) Since the limit equals $f(0, 0) = 0$, the function is **continuous at the origin**.

(c) If f_x and f_y exist in a neighborhood of (a, b) and are continuous at (a, b) , then f is differentiable at (a, b) .

Geometrically, differentiability means the surface has a well-defined tangent plane at that point, and the function is locally well approximated by its linearization.

[CLO #1: Apply multivariable calculus concepts to solve standard mathematical problems involving partial derivatives, multiple integrals, and basic vector fields.]

Question 4

[10 Marks]

(a) In an AI model, the prediction accuracy A depends on three factors: training data size D , model complexity C , and time t . Thus, $A = A(D, C, t)$. Both data size and model complexity can change over time as more data is collected and the model is refined:

$$D = D(t), \quad C = C(t).$$

Using the chain rule, write an expression for the rate of change of accuracy with respect to time, $\frac{dA}{dt}$. (3)

(b) Consider a function $f(x, y)$ for which you know the following values at the point $(1, 2)$:

$$f(1, 2) = 5, \quad f_x(1, 2) = 3, \quad f_y(1, 2) = -1$$

$$f_{xx}(1, 2) = 4, \quad f_{xy}(1, 2) = 2, \quad f_{yy}(1, 2) = 6$$

(i) Find the second-order Taylor expansion of $f(x, y)$ around $(1, 2)$. (2)

(ii) What is the geometric interpretation of a saddle point in terms of curvature? (1)

(c) A machine learning engineer is training a linear regression model with three features. The model's parameters are w_1 , w_2 , and w_3 (weights for features 1, 2, and 3). The loss function J (mean squared error) that the engineer needs to minimize is:

$$J(w_1, w_2, w_3) = w_1^2 + 2w_2^2 + 3w_3^2 - 2w_1w_2 + 4w_1w_3 - 2w_2w_3 - 2w_1 + 4w_2 - 6w_3 + 10$$

You are given that $(w_1, w_2, w_3) = (1, -1, 1)$ is a critical point of J .

Use the determinant test to classify the nature of the critical point as a local minimum, local maximum, or saddle point. (4)

Solution

(a) Since

$$A = A(D, C, t), \quad D = D(t), \quad C = C(t),$$

by the multivariable chain rule,

$$\frac{dA}{dt} = \frac{\partial A}{\partial D} \frac{dD}{dt} + \frac{\partial A}{\partial C} \frac{dC}{dt} + \frac{\partial A}{\partial t}.$$

This gives the total rate of change of accuracy with respect to time.

(b) (i) The second order Taylor expansion of $f(x, y)$ about $(1, 2)$ is

$$\begin{aligned} T_2(x, y) &= f(1, 2) + f_x(1, 2)(x - 1) + f_y(1, 2)(y - 2) \\ &+ \frac{1}{2} [f_{xx}(1, 2)(x - 1)^2 + 2f_{xy}(1, 2)(x - 1)(y - 2) + f_{yy}(1, 2)(y - 2)^2]. \end{aligned}$$

Substituting the given values:

$$\begin{aligned} T_2(x, y) &= 5 + 3(x - 1) - (y - 2) \\ &+ \frac{1}{2} [4(x - 1)^2 + 4(x - 1)(y - 2) + 6(y - 2)^2]. \end{aligned}$$

(ii) A saddle point occurs when the surface curves upward in one direction and downward in another.

Geometrically, it means the point is neither a local maximum nor a local minimum; the curvature changes sign in different directions (like a mountain pass).

(c) The Hessian matrix of J is formed from second partial derivatives.

First compute second derivatives:

$$J_{w_1 w_1} = 2, \quad J_{w_2 w_2} = 4, \quad J_{w_3 w_3} = 6,$$

$$J_{w_1 w_2} = J_{w_2 w_1} = -2,$$

$$J_{w_1 w_3} = J_{w_3 w_1} = 4,$$

$$J_{w_2 w_3} = J_{w_3 w_2} = -2.$$

Thus the Hessian matrix is

$$H = \begin{pmatrix} 2 & -2 & 4 \\ -2 & 4 & -2 \\ 4 & -2 & 6 \end{pmatrix}.$$

Leading principal minors:

$$D_1 = 2 > 0,$$

$$D_2 = \begin{vmatrix} 2 & -2 \\ -2 & 4 \end{vmatrix} = 8 - 4 = 4 > 0,$$

$$D_3 = \det(H).$$

Compute determinant:

$$\begin{aligned}\det(H) &= 2 \begin{vmatrix} 4 & -2 \\ -2 & 6 \end{vmatrix} - (-2) \begin{vmatrix} -2 & -2 \\ 4 & 6 \end{vmatrix} + 4 \begin{vmatrix} -2 & 4 \\ 4 & -2 \end{vmatrix} \\ &= 2(24 - 4) + 2(-12 + 8) + 4(4 - 16). \\ &= 2(20) + 2(-4) + 4(-12). \\ &= 40 - 8 - 48 = -16.\end{aligned}$$

Since $D_3 < 0$, the Hessian is indefinite.

Therefore, the critical point $(1, -1, 1)$ is a **saddle point**.

Good Luck